MIL-C-43806 (GL)

29 August 1972
(see 6.5)

MILITARY SPECIFICATION

CLOTH, FELT: WOOL, PRESSED (JUNIOR R.O.T.C.)

1. SCOPE

1.1 This specification covers the requirements for one type of mechanical-roll felt.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

FEDERAL

PPP-P-1132 - Packaging and Packing of Woolen, Worsted and Wool Blend (Synthetic Fiber; Cotton) Fabrics.

MILITARY

MIL-C-43665 - Cloth Wool: Mothproofing Treatment Of.

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

PUBLICATIONS

Rules and Regulations Under the Wool Products Labeling Act of 1939.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

FSC 8305

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Standard sample.- The finished felt shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 Material.-

3.2.1 Wool.- The wool fibers shall be fleece, pulled wool, wool noil, reprocessed wool, reused wool, or a combination thereof.

3.3 Color.- The color of the finished felt shall be Teal Blue. (Cable No. 65024) and shall be obtained by stock or piece dyeing (see 6.4).

3.3.1 Colormatching.- The color of the finished felt shall match the standard sample under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

3.3.2 Colorfastness.- The finished felt shall show fastness to light and wet drycleaning equal to or better than the standard sample. When no standard sample has been established or designated as applicable to colorfastness, the finished felt shall show "good" fastness to light at 40 standard fading hours and "good" fastness to wet drycleaning. Testing shall be as specified in 4.4.

3.4 Physical requirements.- The finished felt shall conform to the requirements specified in table I when tested as specified in 4.4.

TABLE I.- Physical requirements

Thickness			Weight	
Normal	Minimum	Maximum	Minimum	Maximum
Inch	Inch	Inch	lb/sq.yd.	lb/sq.yd.
0.047	0.040	0.054	0.650	0.750

3.4.1 Width.- The finished felt shall be trimmed and the minimum width after trimming shall be 54 inches.

3.5 Mothproofing.- The felt shall be mothproofed in accordance with MIL-C-43665.

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3.6 Finish and pH.- The felt shall have a chevron type clear hard finish equal to or better than the standard sample. The distortion of the surface of the felt shall be equal to or less than the distortion of the standard sample when both are rubbed by hand. The pH of the water extract of the finished felt shall be not lower than 4.0 nor higher than 8.0 when tested as specified in 4.4.

3.6.1 Relaxation shrinkage.- The finished felt shall be sponged and the relaxation shrinkage after sponging shall not exceed 3.5 percent in length and 2.5 percent in width when tested as specified in 4.4.

3.7 Length.- The felt shall be furnished in rolls of 26 to 55 yards. Each roll shall contain not more than two pieces neither of which shall be less than 3 yards in length.

3.8 Wool content.- The finished felt shall contain not less than 95 percent wool based on the original dry weight of the specimen when tested as specified in 4.4.

3.9 Wool content label.- Felt manufactured under the specification shall be labeled in accordance with the Wool Products Labeling Act of 1939.

3.10 Workmanship.- The finished felt shall conform to the quality established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

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4.2.2. Examination of the end item.- Examination of the end item shall be in accordance with 4.2.2.1 through 4.2.2.3.

4.2.2.1 Examination for local defects.- The felt shall be examined on one side only. However, for each successive roll in the sample, that side shall be alternated. All defects shall be counted regardless of their proximity to each other, except where two or more defects represent a single local condition, in which case only one defect shall be counted. A defect extending over more than 1 continuous linear yard shall be counted as one defect for each such yard, or fraction thereof, in which it occurs. The sample unit shall be one linear yard. The lot size for determining the sample size shall be expressed in units of 1 linear yard each. The inspection level shall be level I of MIL-STD-105. The number of rolls from which the sample yardage is to be selected shall be in accordance with table II herein. An approximate equal number of yards shall be examined from each roll.

DEFECTS

Cut, hole or tear.
Open, breaks, or thin area.
Abrasion or rub.
Wrinkle or crease.
Surface pucker or ripple.
Waviness or inability to lay flat.
High, thick or lumpy area.
Burrs, shives, seed or other naturally occurring foreign matter.
Spots, stains, or streaks not due to fiber distribution.
Jagged or untrimmed edge extending 9 inches or more on a side.
Soft area noticeably less firm than adjoining normal area.
Extraneous foreign matter.
Width less than specified.

4.2.2.1.1 Acceptable quality levels.- The acceptable quality level shall be 10.0 defects per hundred units.

4.2.2.2 Examination for overall defects.- Each defect listed shall be counted not more than once in each roll examined. The sample unit shall be one roll. The number of rolls for this examination shall be in accordance with table II. The lot shall be unacceptable if the defects found exceed the maximum number permitted in table II.

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OVERALL DEFECTS

Wrong color, i.e., not as specified.
Degree and character of finish not equal to standard sample 1/.
Off shade, shaded, mottled.
Rancid or unclean odor.
Ticket or tag not attached.
Marking incorrect, incomplete, or illegible.
Not labeled in accordance with the Wool Products Labeling Act of 1939.

1/ The distortion of the surface shall be determined by rubbing by hand with 5 rubs of equal hand pressure.

4.2.2.3 Examination for length.-

4.2.2.3.1 Individual rolls.- The rolls shall be examined for defects listed below. The sample unit for this examination shall be one roll. The number of rolls selected as the sample (sample size) for this examination and the acceptance number shall be in accordance with table II.

LENGTH DEFECTS

Any gross length less than the minimum specified or more than 2 yards less than the gross length marked on the ticket.
Any gross length more than the maximum specified.
Any roll containing more than two pieces.
Any piece less than 3 yards.

4.2.2.3.2 Examination for total.- The lot shall be unacceptable if the total of the actual gross lengths of the rolls in the sample is less than the total of the gross lengths marked on the roll tags. The number of rolls or containers selected as a sample shall be those taken from the overall examination.

TABLE II

Lot size in yards	Sample size in rolls	Acceptance number
Up to 1200 1/	3	0
1201 up to and including 3,200	5	0
3,201 up to and including 10,000	7	0
10,001 up to and including 35,000	10	0
35,001 and over	15	1

1/ If lot contains fewer than three rolls, each roll in the lot shall be examined.

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4.3 Examination of preparation for delivery requirements.- An examination shall be made to determine that packaging, packing and marking comply with the Section 5 requirements.

4.4 Testing of the end item.- The test methods listed in table III shall be in accordance with FED-STD-191 except as otherwise noted, and as described herein. The physical and chemical values specified in section 3 apply to the average number of test specimens for a sample unit for test purposes, as specified in applicable test methods. The sample unit shall be 2-1/2 continuous yards full width of the finished felt. The lot shall be unacceptable if one or more sample units fail to meet any test requirement specified. The lot size shall be expressed in units of one yard. All test reports shall contain the individual values utilized in expressing the final results. The sample size (number of sample units) shall be in accordance with the following:

Lot size (yards)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE III.- Test methods

Characteristics	Requirement paragraph	Test method
Colorfastness to:		
Light	3.3.2	5660
Wet drycleaning	3.3.2	5622
Thickness	3.4	5030
Weight	3.4	5041
Mothproofing content	3.5	1/
pH	3.6	2811
Relaxation shrinkage		
Length	3.6.1	5558
Width	3.6.1	5558
Wool content	3.8	2101

1/ As specified in MIL-C-43665.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C.- The felt, put up as specified, shall be packaged in accordance with the applicable requirements of PPP-P-1132.

5.2 Packing.- Packing shall be level A, B or C as specified (see 6.2).

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5.2.1 Levels A, B and C.- The felt shall be packed in accordance with the applicable requirements of PPP-P-1132.

5.3 Marking.- In addition to any special marking required by the contract or order, shipments shall be marked in accordance with PPP-P-1132.

6. NOTES

6.1 Intended use.- The wool felt is to be used as facing on the lapels of uniform for the Junior R. O. T. C.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Standard sample.- For access to standard sample, address the procuring office issuing the invitation for bids.

6.4 The standard for Teal Blue has been dyed with the formulation as shown below. This is suggested, but not mandatory.

Acid Orange 86
Neutral Premetallized Blue GL
Neutral Premetallized Green GL

6.5 Supersession data.- This document includes the requirements of Natick Laboratories Purchase Description Number 300A dated October 1965.

Custodian:

Army - GL

Review activity:

Army - MD

Preparing activity:

Army - GL

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R355

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

Cloth, Felt: Wool, Pressed (Junior R.O.T.C.) MIL-C-43806(GL)

ORGANIZATION

CITY AND STATE	CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both inan envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*	DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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